

THE LAST VARIABLE

A Middle-School Math Escape Game



Print · Cut · Play | 7th Grade (7.NS · 7.EE · 7.RP · 7.G) | Differentiated for inclusion classrooms

How to Play

The school's mainframe is collapsing. A rogue program — **The Glitch** — is deleting everything inside it, one variable at a time. A team of **Circuit Breakers** is trapped. Bring three generators back online and reach the exit before the Glitch deletes you all — before one of you becomes the last variable.

Players & Roles

4–5 players. One is **The Glitch** (the hunter, solo). The rest are **Circuit Breakers** (the team). Deal the Glitch role with the rotation cards — it passes every game, and everyone plays it before anyone repeats.

Setup (5 min)

1. Lay out the board; place 3 generator markers and the exit/start tokens.
2. Deal roles (rotation cards).
3. Each Breaker picks 1 of 8 character cards (no repeats); the Glitch picks 1 of 3.
4. Each player has a **difficulty tier (1–3)** — you assign it or they choose. The Glitch defaults to Tier 3 (a struggling student may drop to Tier 2).
5. All tokens start on the Start node. Set the turn track to **10**. (5 players? The Glitch starts with +1 move.)

Turns

Breaker turn: move up to 3 nodes → (optional) activate power → interact (generator / door / teammate).

Glitch turn: move up to 4 nodes (**+1 per generator already online**) → (optional) power. If the Glitch shares a node with a Breaker, that Breaker is **frozen** — locked for 1 full round, then revivable.

Powers — the engine

To fire any power: draw a problem card of its skill **at your tier**, solve it, check the key. **Correct** → **power fires**. **Wrong** → **turn ends**. A Tier-1 and a Tier-3 player fire the same power — they just pull different cards. Most powers have a cooldown. **Solve at Tier 3 and that cooldown drops by 1 (min 0)** — harder math lets you use your power more often.

Generators & Winning

Each generator needs TWO Breakers — each solves a card from the node's deck at their tier (one holds half the code). No soloing the objectives. **Rubber-band:** every generator that comes online gives the Glitch +1 move; once all 3 are online, freezes last 2 rounds.

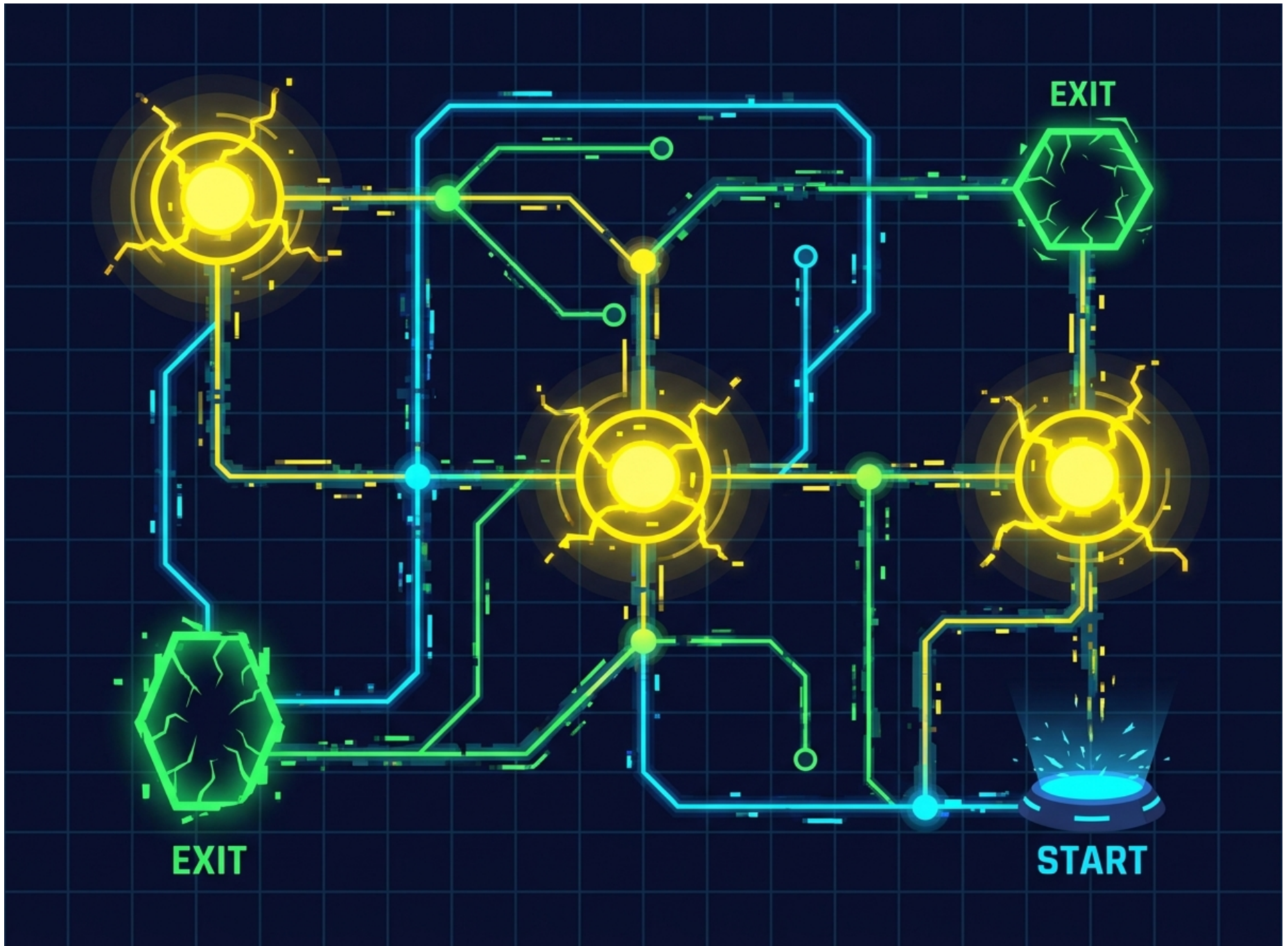
Circuit Breakers win when all 3 generators are online AND a Breaker reaches any exit before the turn track runs out.

The Glitch wins if the track runs out, or every Breaker is frozen at once. Revive a frozen Breaker (after its 1-round lockout) with The Medic (within 2 nodes), or by moving onto their node and solving a Percentages card at your tier.

	Breakers	The Glitch
Players	3–4	1
Move / turn	3 nodes	4 nodes, +1 per generator online
Problem tier	assigned/choice 1–3; T3 = –1 cooldown	Tier 3 default (timed), may drop to T2

Generators	2 Breakers each solve to bring 1 online	—
Win by	all 3 generators + reach any exit	run out clock OR freeze all

The Mainframe — Game Board



Yellow bursts = generator nodes (objectives). Green hexes = exit nodes. Blue platform = start. Dots = path nodes; move along the lines.

Circuit Breakers — pick 1 of 8



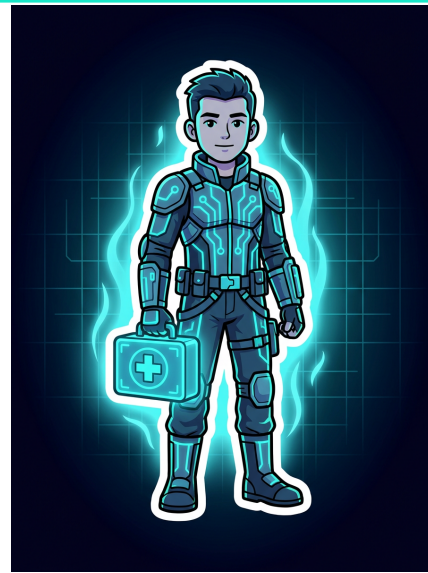
The Hacker

Override

Bring one generator online or open one exit from an adjacent node.

Solve: Algebra | Cooldown: 2 rounds

"There's always a backdoor."



The Medic

Reboot

Revive a frozen teammate within 2 nodes (after its 1-round lockout).

Solve: Percentages | Cooldown: 2 rounds

"Stay with me — you're not deleted yet."



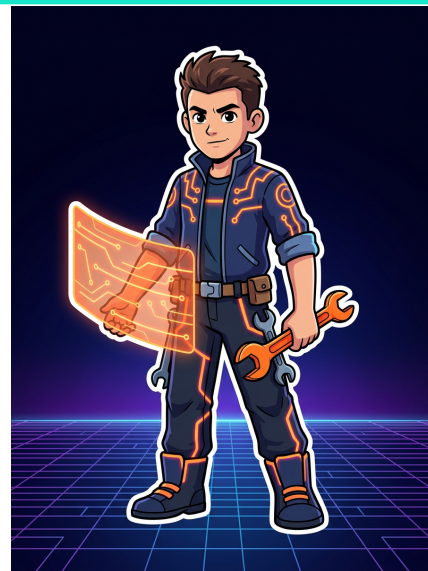
The Scout

Ping

Reveal 3 hidden tiles and the Glitch's location this turn.

Solve: Geometry | Cooldown: 1 round

"I see everything in here."



The Engineer

Barricade

Jam a door or corridor for 2 rounds. The Glitch can't pass.

Solve: Fractions | Cooldown: 2 rounds

"Try getting through THAT."

Circuit Breakers (cont.)



The Runner

Dash

Move +3 extra nodes this turn (6 total).

Solve: Mixed (quick) | Cooldown: 1 round

"Catch me."



The Decoder

Translate

Read one teammate's split-info clue to the table.
Once per game.

Solve: Any deck | Cooldown: 1× game

"Hand me that — I'll read it."



The Guardian

Firewall

Block one freeze against yourself or an adjacent teammate.

Solve: Fractions | Cooldown: 2 rounds

"Not on my watch."



The Trickster

Decoy

Drop a fake signal; the Glitch must move toward it next turn.

Solve: Geometry | Cooldown: 2 rounds

"Look over there. No — over THERE."

The Glitch — pick 1 of 3



The Corruptor

Lockdown

Seal one exit or freeze one tile for 2 rounds.

Solve: Algebra (T3, timed) | Cooldown: 2 rounds

"This door is mine now."



The Stalker

Trace

Reveal the nearest Breaker and move +2 toward them now.

Solve: Geometry (T3, timed) | Cooldown: 1 round

"I can smell the math on you."



The Overclock

Surge

Take one extra full turn right now.

Solve: Mixed (T3, timed) | Cooldown: 2 rounds

"Faster. Faster. FASTER."

Fractions Deck — cut-apart cards

FRACTIONS · TIER 1

$$1/4 + 2/4 = ?$$

FRACTIONS · TIER 1

$$5/8 - 2/8 = ?$$

FRACTIONS · TIER 1

$$2/5 + 1/5 = ?$$

FRACTIONS · TIER 1

$$1/3 \times 1/2 = ?$$

FRACTIONS · TIER 1

$$7/10 - 3/10 = ?$$

FRACTIONS · TIER 1

$$3/7 + 2/7 = ?$$

FRACTIONS · TIER 2

$$1/2 + 1/3 = ?$$

FRACTIONS · TIER 2

$$3/4 - 1/6 = ?$$

FRACTIONS · TIER 2

$$2/3 \times 3/5 = ?$$

FRACTIONS · TIER 2

$$1/2 \div 1/4 = ?$$

FRACTIONS · TIER 2

$$5/6 - 1/4 = ?$$

FRACTIONS · TIER 2

$$2/3 + 3/4 = ?$$

FRACTIONS · TIER 3

$$2 \frac{1}{2} + 1 \frac{3}{4} = ?$$

FRACTIONS · TIER 3

$$3 \frac{1}{3} - 1 \frac{5}{6} = ?$$

FRACTIONS · TIER 3

$$2 \frac{2}{3} \times 1 \frac{1}{2} = ?$$

FRACTIONS · TIER 3

$$3 \frac{3}{4} \div 1 \frac{1}{2} = ?$$

FRACTIONS · TIER 3

$$4/5 \times 2 \frac{1}{2} = ?$$

FRACTIONS · TIER 3

$$5 \frac{1}{6} - 2 \frac{3}{4} = ?$$

Algebra Deck — cut-apart cards

ALGEBRA · TIER 1

$$x + 7 = 12$$

ALGEBRA · TIER 1

$$x - 4 = 9$$

ALGEBRA · TIER 1

$$3x = 21$$

ALGEBRA · TIER 1

$$x/2 = 6$$

ALGEBRA · TIER 1

$$x + 15 = 20$$

ALGEBRA · TIER 1

$$5x = 45$$

ALGEBRA · TIER 2

$$2x + 3 = 11$$

ALGEBRA · TIER 2

$$3x - 5 = 16$$

ALGEBRA · TIER 2

$$x/4 + 2 = 6$$

ALGEBRA · TIER 2

$$5x + 7 = 32$$

ALGEBRA · TIER 2

$$4x - 9 = 7$$

ALGEBRA · TIER 2

$$2x - 8 = -2$$

ALGEBRA · TIER 3

$$3(x + 2) = 21$$

ALGEBRA · TIER 3

$$2(x - 4) + 3 = 11$$

ALGEBRA · TIER 3

$$5x - 3 = 2x + 12$$

ALGEBRA · TIER 3

$$4(x + 1) = 2x + 14$$

ALGEBRA · TIER 3

$$6x + 2 = 4x + 18$$

ALGEBRA · TIER 3

$$3x + 7 = 5x - 9$$

Geometry Deck — cut-apart cards

GEOMETRY · TIER 1

Rectangle 8×3 : area?

GEOMETRY · TIER 1

Rectangle 8×3 :
perimeter?

GEOMETRY · TIER 1

Triangle $b10\ h4$: area?

GEOMETRY · TIER 1

Square side 6: area?

GEOMETRY · TIER 1

Square side 6:
perimeter?

GEOMETRY · TIER 1

Triangle $b8\ h5$: area?

GEOMETRY · TIER 2

Circle $r5$: area?
($\pi=3.14$)

GEOMETRY · TIER 2

Circle $r5$:
circumference?

GEOMETRY · TIER 2

Parallelogram $b12\ h4$:
area?

GEOMETRY · TIER 2

Circle $r10$: area?

GEOMETRY · TIER 2

Circle $d14$:
circumference?

GEOMETRY · TIER 2

Trapezoid $b6, b10\ h4$:
area?

GEOMETRY · TIER 3

Prism $4 \times 3 \times 5$: volume?

GEOMETRY · TIER 3

Cube side 4: surface
area?

GEOMETRY · TIER 3

Cylinder $r3\ h10$:
volume?

GEOMETRY · TIER 3

Tri-prism ($b6\ h4$)
 $len10$: volume?

GEOMETRY · TIER 3

Prism $5 \times 3 \times 2$: surface
area?

GEOMETRY · TIER 3

Cylinder $r5\ h4$:
volume?

Percentages Deck — cut-apart cards

PERCENTAGES • TIER 1

10% of 50

PERCENTAGES • TIER 1

25% of 80

PERCENTAGES • TIER 1

50% of 24

PERCENTAGES • TIER 1

20% of 60

PERCENTAGES • TIER 1

10% of 200

PERCENTAGES • TIER 1

75% of 40

PERCENTAGES • TIER 2

15% of 60

PERCENTAGES • TIER 2

30% of 90

PERCENTAGES • TIER 2

12 is what % of 48?

PERCENTAGES • TIER 2

40% of 75

PERCENTAGES • TIER 2

12% of 150

PERCENTAGES • TIER 2

18 is what % of 30?

PERCENTAGES • TIER 3

\$80 item, 25% off:
price?

PERCENTAGES • TIER 3

\$50 item, 8% tax:
total?

PERCENTAGES • TIER 3

40→50: % increase?

PERCENTAGES • TIER 3

\$120 item, 15% off:
price?

PERCENTAGES • TIER 3

60→45: % decrease?

PERCENTAGES • TIER 3

\$200 item, 6% tax:
total?

Answer Keys

Fractions

Tier 1: 1) $\frac{3}{4}$ 2) $\frac{3}{8}$ 3) $\frac{3}{5}$ 4) $\frac{1}{6}$ 5) $\frac{2}{5}$ 6) $\frac{5}{7}$

Tier 2: 1) $\frac{5}{6}$ 2) $\frac{7}{12}$ 3) $\frac{2}{5}$ 4) 2 5) $\frac{7}{12}$ 6) $1\frac{5}{12}$

Tier 3: 1) $4\frac{1}{4}$ 2) $1\frac{1}{2}$ 3) 4 4) $2\frac{1}{2}$ 5) 2 6) $2\frac{5}{12}$

Algebra

Tier 1: 1) 5 2) 13 3) 7 4) 12 5) 5 6) 9

Tier 2: 1) 4 2) 7 3) 16 4) 5 5) 4 6) 3

Tier 3: 1) 5 2) 8 3) 5 4) 5 5) 8 6) 8

Geometry

Tier 1: 1) 24 2) 22 3) 20 4) 36 5) 24 6) 20

Tier 2: 1) 78.5 2) 31.4 3) 48 4) 314 5) 43.96 6) 32

Tier 3: 1) 60 2) 96 3) 282.6 4) 120 5) 62 6) 314

Percentages

Tier 1: 1) 5 2) 20 3) 12 4) 12 5) 20 6) 30

Tier 2: 1) 9 2) 27 3) 25% 4) 30 5) 18 6) 60%

Tier 3: 1) \$60 2) \$54 3) 25% 4) \$102 5) 25% 6) \$212

Token Sheet — cut out & play

Sample player tokens (use any character's head crop) and marker tokens. Print on cardstock.

